

BEM-C302
SEMESTER EXAMINATION- DECEMBER 2023
CLASS: B.TECH SEMESTER: III
ENGINEERING MATHEMATICS-III

TIME: 3 HOUR

MAX. MARKS: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each Q carries six marks.

(5 x 6 = 30 Marks)

Q1. Evaluate the Laplace transform of $\int_0^t te^{-4t} \sin 3t dt$.

Q2. Express the following function in terms of unit step function and determine its Laplace transform

$$f(t) = \begin{cases} \cos t & 0 < t < \pi \\ \cos 2t & \pi < t < 2\pi. \\ \cos 3t & t > 2\pi \end{cases}$$

Q3. Find inverse Laplace transform of $\log\left(1 + \frac{1}{s^2}\right)$.

Q4. Express the function

$$f(x) = \begin{cases} \sin x & 0 \leq x \leq \pi \\ 0 & x \geq \pi \end{cases}$$

as a Fourier sine integral and deduce that $\int_0^\infty \frac{\sin \lambda x \sin \pi \lambda}{1 - \lambda^2} d\lambda = \frac{\pi}{2} \sin x$, $0 \leq x \leq \pi$.

Q5. Apply Parseval's identity to show that $\int_0^\infty \frac{\sin at}{t(a^2 + t^2)} dt = \frac{\pi}{2} \left(\frac{1 - e^{-a^2}}{a^2} \right)$.

Q6. Show that $Z \frac{1}{n!} = e^{1/z}$ hence evaluate $Z \frac{1}{(n+1)!}$ and $Z \frac{1}{(n+2)!}$

Q7. Test that the function $u(x, y) = e^x \cos y$ is harmonic. Construct the corresponding analytic function $f(z) = u + iv$.

Q8. Evaluate $\oint_C \frac{z^2}{z(z-1)(z-3)}$ where (i) $C: |z+2| = 1$ (ii) $C: |z| = 2$.

Q9. Apply iteration method to approximate the real root of the equation $3x = \cos x + 1$, correct to 3 decimal places.

Q10. Apply Newton Raphson method to approximate the real root of $xe^x - 1 = 0$ correct to 4 decimal places.

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any *four* questions in detail. Each question carries 10 marks

(4 x 10 = 40 Marks)

Q1. Employ Laplace transform to solve the following differential equation

$$y'' + 2y' + y = te^{-t}, \quad y(0) = 1, y'(0) = -2.$$

Q2. Solve the heat equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$ subject to conditions

(i) $u = 0$ when $x = 0, t > 0$

(ii) $\frac{\partial u}{\partial x}(0, t) = -a(\text{constant})$

(iii) $u(x, t)$ is bounded.

Q3. Evaluate $f(x)$ if its Fourier sine transform is $\frac{1}{\lambda} e^{-a\lambda}$. Hence deduce $F_s^{-1}\left(\frac{1}{\lambda}\right)$.

Q4. Apply Z transform to solve the difference equation $y_{n+2} + 6y_{n+1} + 9y_n = 4^n$,

$$\text{where } y(0) = 0, y(1) = 0.$$

Q5. State the initial value theorem and determine f_2 and f_3 where $f(z) = \frac{2z^2 + 3z + 14}{(z-3)^3}$

Q6. Show that the function $f(z) = u + iv$, where

$$f(z) = \begin{cases} \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2} & z \neq 0 \\ 0 & z = 0 \end{cases}$$

Satisfies the Cauchy Riemann equations at $z=0$. Examine the analyticity of function at $z=0$. Justify your answer?

Q7. Using Contour integration technique evaluate the real integral $\int_0^{2\pi} \frac{\cos 3\theta}{5 - 4\cos\theta} d\theta$.

Q8. Compute the real root of $3x + \sin x - e^x = 0$ using secant method correct to four decimal places.